BEVY_ZEROVERSE: AUDITABLE PROCEDURAL INDOOR DATA FOR MULTI-VIEW RECONSTRUCTION

A PREPRINT

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ABSTRACT

We describe an open-source synthetic data engine combining primitive-object scenes with furnished procedural interiors, calibrated multi-view cameras, optional human motion, and aligned geometric annotations. Indoor generation samples continuous architectural, furnishing, appearance, lighting and camera parameters under geometric constraints. A continuous camera-baseline control coordinates spacing, shared-surface overlap and path variation. The architectural evaluation audits 512 consecutive rooms and 256 aligned views across 32 rooms. A separately identified camera-baseline reference study covers 2,048 layouts and 512 rendered rooms, with co-visibility measurements at five baseline values. Current capabilities and measured cohorts are distinguished, with exact configurations and reproducible artifacts. Photographic realism, unlimited-process stability and downstream pretraining gains are not established by these diagnostics.



Figure 1: Current generated architecture: mezzanine stairs, a sunken lounge and a raised platform with an arched portal. Native RGB at $t = 0$; seeds 7, 8 and 6, cameras 3, 3 and 1. Plans, calibrated views, aligned annotations and the unfiltered cohort follow in the main evaluation.

1 Purpose and scope

Sparse-view reconstruction benefits from observations of shared surfaces: cameras that see unrelated objects offer little direct correspondence supervision. LRM-Zero demonstrates the potential of procedurally synthesized object data [1]; MVSplat uses cross-view matching to estimate geometry [2]. Our contribution is an inspectable data-generation system and its bounded evaluation, rather than a trained reconstruction model or a reconstruction accuracy result.

The engine retains the existing Object, CornellCube, Room, SemanticRoom and Human scene modes, and exposes furnished interiors separately as `ProceduralIndoor`. The runtime uses Bevy 0.20, optional Burn 0.22 inference, and published AnnyBody/ARDY integrations. Office, conference, lounge, classroom, library and related activity priors influence a compositional program; they do not select complete meshes or a finite collection of room templates.

Most architecture, furniture, plants and clutter require neither mesh repositories nor photographic texture collections. AnnyBody human meshes and optional motion/embedding models are explicit pretrained dependencies.

2 Procedural scene program

A scene is generated as $S = G(s, \theta)$, where s seeds independent random streams and θ contains user-controlled densities, activity, quality, camera and motion policies. The serialized manifest records the sampled program, object transforms and seeds, people, photometry and camera paths. A separate generator version and capture-engine identity invalidate incompatible resumptions. Determinism means reproducing the same program and settings with compatible assets and engine versions; bit-identical rasterization across GPU vendors is not promised.

2.1 Architecture, objects and appearance

Room area and aspect, functional zones, aisle widths and activity mixtures are sampled continuously. The exterior footprint supports taper, chamfered corners and a concave cut-in. A shared architectural program controls polygon triangulation, wall apertures, sloping ceilings, arched portals, pillars, floor levels and furnished mezzanines with stairs and guards. Furniture support, camera clearance, fixture anchors and voxel export use the same surfaces. Windows vary aperture dimensions, sill/head heights, bay widths, frames, inset depth and shades across multiple exterior edges. An internal glazed connection retains a furnished neighboring room. This is a bounded architectural grammar: courtyard cut-ins remain on the rear facade, neighboring rooms and mezzanine decks remain rectangular, and arbitrary multi-storey connectivity, structural analysis and building-code compliance are outside scope.

Furniture is assembled from material-specific parts. Tables vary outline, taper, thickness and leg/pedestal construction; chairs vary seat shell, curvature, back proportions, recline, arms, base and headrest. Sectional seating, shelving, potted plants, outlets, switches, ceiling lights, whiteboards, displays and tabletop objects extend the semantic inventory. Placement uses support relationships, clearance checks, zone-dependent furnishing fields and bounded rejection. Chairs sample partial and deep insertion beneath tables, including occupied seats; the rendered leg, pedestal, brace and panel program also supplies collision geometry. Working poses mix free reaches with one- or two-palm tabletop support using fixed-length arm IK and skin-aware contact fitting. This is static surface support, not simulated grasping or animated contact dynamics. Surface clutter varies shared arrangement, handedness, spacing and local jitter. Object yaw and arrangement are sampled rather than restricted to orthogonal grids. Collision proxies approximate detailed meshes; geometric validity is checked separately from physical plausibility.

Surfaces use 35 role-specific base material slots with procedural albedo, normal and roughness maps, metric UV scales and mip chains. Recipes vary spatial frequency, directional grain, textile structure, mineral veining, relief, wear, color and roughness. Upholstery finish slots share three independently seeded structure programs per substrate; clothing reuses these maps with independent actor pigmentation and atlas scaling. Woven textiles sample over/under grammars, yarn diameter, slub, crimp, dyed warp/weft and fuzz. Leather varies grain spacing, stretch, pore relief, nap, pigmentation and polish. Coupled albedo, metric normal, cavity and roughness fields are baked into mipmapped textures. Textile directional GGX scattering approximates aggregate fibre reflection; it is not a measured cloth multiple-scattering BSDF. Mineral programs blend cement binder, exposed aggregate, pinholes and warped multiscale vein deposits with independent polish. Wall finishes vary roller stipple, sprayed and flattened plaster relief, trowel marks and gloss; glazed ceramics add restrained crackle and a protective lobe. Timber separates grain cuts from stain absorption, bleaching, pore fill and varnish; face and edge grain share their finish. Foliage atlases keep veins aligned to the blade fold and vary chlorophyll patches, pale variegation, wax and thin-surface transmission. Timber and foliage finish slots share two texture structures per role. Timber and stone floor atlases use 512-square maps to resolve separately cut boards and stone tiles; other substrates use 256-square maps. Screens, boards and printed pages have independently seeded content. Only referenced finish variants are built during runtime preparation. Glass recipes vary roughness, refractive index, tint, attenuation and thickness. Direct lights vary placement, intensity, color temperature, circuit activity and direction; exposure, daylight and sky radiance are independently recorded. Low-light samples are retained in evaluation.

Timber samples offset and tapered growth-ring cuts, uneven earlywood/latewood, elongated pores and sparse branch knots; floor planks have independent cuts. Unresolved normal variation contributes to filtered GGX roughness. Chrome retains a conductive base layer with continuously varied polished/brushed roughness and anisotropy; micron scratches are represented as scattering rather than quantized visible bumps. A static scene-derived HDR environment traces the rendered triangle assemblies and visibility-tested direct lights using the shared transport BVH. Linear radiance is stored in float16 cubemaps, with GGX specular and cosine diffuse convolution; source footprints suppress sparse-emitter filtering noise. Secondary bounce energy is approximate. One reflection origin does not provide per-object parallax, live moving-human reflections or a full specular path-tracing reference.

People retain AnnyBody geometry with continuous phenotype/pose parameters, fitted garments, surface hair and facial details. Static people remain the default. Opt-in ARDY motion combines seeded procedural text sequences with object-aware waypoint planning and scene-time playback. Requesting static scenes does not initialize motion inference. Motion models use the upstream loaders and caches; cached prompt text does not replace the motion seed. Clothing, hair, joint limits and collision handling are approximations, particularly during complex motion.

2.2 Multi-view sampling and training defaults

Camera centers and complete trajectories are constrained to the primary furnished room by default. The sampler varies focal length, position, target, roll and path shape, checks obstacle clearance and prefers visible content. Static capture uses one timestep; optional trajectories support repeated calibrated observations. A reference camera anchors a connected set of views. For each other camera and five trajectory times, we require

$$\min\{o_{0\to i}(t), o_{i\to 0}(t)\} \geq \tau, \qquad b_{\min} \leq \|c_i(t) - c_0(t)\| \leq b_{\max},$$

where o estimates first-surface overlap using visibility rays and object/architecture proxies. Overlap constrains reference edges. An additional minimum separation applies to every camera pair at 33 synchronized path samples. Group spread is the horizontal camera-center covariance's minor/major standard-deviation ratio. Moving pairs must differ in normalized RMS displacement after removing their starting offsets. Full-path swept collision checks remain; sampled spread and overlap are not continuous-time or rendered-pixel guarantees.

The camera-baseline control $b \in [0, 1]$ expands into explicit advanced settings: minimum pair separation $0.04 + 0.46b$ metres, minimum reference distance $0.06 + 2.30b$ metres, maximum reference distance $0.35 + 9.65b$ metres, overlap target $0.65 - 0.55b^{0.6}$, group spread $0.20 + 0.10b$, and independent path variation b. The default is $b = 0.5$. Wide settings permit distant viewpoints and lower overlap; narrow settings favor close views of shared content. Collision and visibility rejection affect the realized distribution. Arbitrary room, camera-count and constraint combinations need not be feasible.

The default grouped sampler remains biased toward a shared fixation region: peer targets inherit the reference target with only ± 0.35 m horizontal and ± 0.15 m vertical jitter. Their endpoint targets inherit the same offset. Baseline changes positional spacing and path variation but does not broaden this target jitter. Positional diversity and overlap checks therefore do not establish independent gaze or convergence diversity.

The inspector exposes this slider above the advanced controls. Any advanced edit switches to explicit custom values. Changes take effect on regeneration, so dragging a control does not rebuild the room. JSON input accepts `{"baseline":0.75}` within the `indoor_camera` configuration. Serialization records the expanded values, avoiding hidden preset precedence. Camera travel length and long-route frequency are separate from spacing between cameras. Joint collision and overlap rejection can favor shorter accepted routes at wide spacing; the minimum travel bound remains explicit. The finite search budget reports impossible settings without silently weakening them.

Indoor training configuration	Default and rationale
Capture cameras / timesteps	Four / one; synchronized views of static geometry
Shared-view policy	$b = 0.5$: $\tau \approx 0.287$, pair separation ≥ 0.27 m, reference distance 1.21–5.175 m; explicit `multiview:null` enables independent views
Group diversity	Horizontal spread ratio ≥ 0.25 for 3+ views; moving-pair normalized RMS difference ≥ 0.075
Room/activity/density	Primary room, Mixed activity, furnishing density 0.65, human density 0.25
Camera path policy	0.03–8 m allowed path length, 0.45 long-path proposal fraction; collision rejection still applies
Human motion	Disabled; static people remain, no ARDY initialization
Native lighting	Auto quality, shadows and approximate baked diffuse GI, 256 rays/probe
Dataset CLI workers / chunks	One process, 16 rooms/chunk; explicit overrides retained
Finite worker lifetime	256 scenes/child for finite indoor jobs with process workers
Default export	RGB and calibration; geometric channels are opt-in

Table 1: Defaults conditional on the procedural-indoor dataset CLI scene. Other scene modes retain their defaults. These are an engineering starting point, not an optimum established by downstream training. Library callers explicitly choose camera and worker counts.

3 Capture and annotation contract

Capture begins only after scene preparation, assets, shader pipelines, lighting and requested human trajectories are ready. A bounded asynchronous readback protocol associates outputs with the requested scene, camera, time and mode; it does not infer readiness from an arbitrary number of frames. Sequential native indoor streams automatically prepare future rooms on a shared bounded CPU pool and can upload one future room's immutable assets during current readback. Future entities, lighting dispatches and motion inference remain gated by scene promotion. This scheduling retains full native quality rather than reducing shadows or geometry. Geometric channels share an RGBA32Float ground-truth raster attachment. RGB uses Bevy's PBR pipeline, shadow maps, tone mapping and optional indirect-lighting approximation. Float32 storage does not imply that all RGB or lighting computations have float32 intermediate precision: HDR color and irradiance volumes include float16 stages.

Output	Meaning and qualification
Color	Display RGB; lossless float storage or explicitly chosen codecs. Separate linear-HDR export supports renderer comparisons.
Depth	Colorized, normalized or metric linear depth; axial/ray conventions; calibration records the selection.
Normal / position	View-space normals encoded as $(n + 1)/2$; world positions affine-normalized by the exported AABB, without clamping. First surface, float32 geometric path.
Semantic	Shared class vocabulary, aligned with depth/normal/position, including ceiling light fixtures.
Optical flow	Forward displacement $(\Delta u, \Delta v)$ in pixels, with correspondence-valid and target-visible masks. Terminal time has no forward correspondence.
Motion vectors	The same exported displacement divided by image width/height. Viewer velocity colors are previews, not these numeric annotations.
Co-visibility	Optional same-time, per-pixel membership in other capture cameras. A uint16 mask assigns one bit per camera (up to 16), with a separate source-valid mask and a consistent additive RGB legend; the source camera and editor are excluded.
Calibration / instances	Camera intrinsics, poses and normalized trajectory time, scene manifest, semantic instance oriented boxes, person joints, and primary-room annotation AABB.
O-Voxel	Primary-room architecture/objects only; excludes neighboring/outside geometry. Requires one timestep and disabled human motion; incompatible requests are rejected.

Table 2: Eight currently supported render modes plus structured annotations. The current architectural cohort exports six modes, including production co-visibility. The separately identified reference gallery exports all eight modes, and its population study derives co-visibility independently from depth and calibration. Glass is a first geometric surface: annotations do not follow refracted or reflected light paths. Flow transports surface correspondence, including supported human deformation, rather than estimating photometric image motion.

Headless generation supports chunked safetensors or per-sample files, compression, resumable identity checks, metrics and bounded worker replacement. RGB and numeric channels have separate encoding contracts. Native generation performs numeric temporal export and O-Voxel export. The WebGPU viewer supports indoor construction, inspection and optional motion inference, using the same manifest and geometry modules; Portable quality reduces unsupported or costly rendering features. Browser velocity visualization is not a qualification of native dataset flow export. The current Wasm path omits native baked diffuse GI and uses its platform lighting fallback.

4 Current architectural evaluation

4.1 Protocol and architectural audit

This cohort was captured with bevy_zeroverse 0.37.2, generator 35, on 512 consecutive seeds (0–511), with 4 cameras per room, furnishing density 0.65 and static human density 0.25. Every audited room passed the recorded layout, support and camera-clearance checks. The first 32 rooms were rendered without appearance filtering at 640×400, with 2 trajectory endpoints: 256 camera/time identities in six aligned modes. The audit contains 512 unique structural signatures after centimetre quantization; this bounded diagnostic does not estimate the total procedural space or learning utility.

Built feature	Rooms	Fraction
Arched portals	194 / 512	37.9%
Chamfered corners	307 / 512	60.0%
Exterior cut-in	221 / 512	43.2%
Interior pillars	425 / 512	83.0%
Mezzanine	46 / 512	9.0%
Oblique walls	458 / 512	89.5%
Raised floor	90 / 512	17.6%
Sloping ceiling	354 / 512	69.1%
Sunken floor	103 / 512	20.1%

Table 3: Architectural features in the complete consecutive population. Features co-occur; fractions do not sum to 100%.

4.2 Geometry, calibration and displayed captures

Footprint areas span 35.05–262.58 m^2, with roof pitch 0.00–11.32°. Figure 2 and Figure 3 pair native RGB with the actual serialized envelope plans and sections. Camera arrows indicate direction, rather than frusta. Solid furnishings are footprint proxies; the sections show roof and floor levels and omit furnishings, pillars and partitions. Structural feature examples are selected independently of rendered appearance; all rooms remain available in the contact sheet and explorer. Metric constraints are not building-code certification.

The page presents all 32 rooms, all four cameras and both endpoints in one explorer, with 1536 channel images. RGB, depth, normals, semantics, positions, co-visibility, camera calibration and plans share the same room/time identity. RGB preserves renderer tone mapping and sRGB encoding without per-image exposure correction or cropping. All previews use lossless WebP. Depth is an 8-bit axial-depth preview divided by 15 m and clamped; normals store $(n + 1)/2$, and positions are affine-normalized by the annotation AABB and clamped. Semantic colors are preserved exactly. These previews are not float32 training labels. Geometric annotations use the first surface, including opaque glass, rather than reflected or refracted optical content.

The native capture protocol uses Auto quality, shadows, SSAO and approximate diffuse GI with 1024 primary rays per probe. Renderer identity and capabilities are recorded in every calibration download. All captured views passed the recorded alignment checks: maximum per-view depth/position p99 error 1.907 μm, reprojection p99 0.00064 pixels and normal-length error 2.980e-7. Every view contains at least 8 semantic classes. 19/19 rooms with humans in the camera's primary functional zone show person pixels; 3/23 rooms with humans anywhere in the main envelope never show person pixels. These are class-level observations, rather than instance-level visibility guarantees.

Scene construction through capture/export takes 1.25–4.45 seconds per room (median 2.46), excluding application startup. This is a bounded capture run, not a sustained-throughput, unlimited-process memory-stability or photographic-realism qualification.

4.3 Co-visibility in every captured room

Production co-visibility accompanies every RGB and geometry view in this cohort. Each valid source pixel stores an exact uint16 membership mask of other capture cameras that see its first surface, with the source bit excluded. One deterministic additive legend is shared across all views. The unified explorer supports all memberships and individual-peer overlays; the selected peer's own tile stays RGB.

Across every room and endpoint, 83.3% of valid source-pixel observations are visible to at least one peer. This pixel-weighted statistic pools 65535995 valid source-pixel observations; it does not count unique 3D points. Exact downloads include 16-bit grayscale membership PNGs, separate 8-bit 0/1 validity PNGs, calibrated transforms and the ordered camera legend. Zero membership with valid=1 is an unshared surface; zero membership with valid=0 is background. Co-visibility and other annotations were captured together, rather than joined from separate runs.

4.4 Matched camera and appearance qualification

A separate prespecified factor sweep covers 2 geometry families, eleven settings, two cameras and two timestamps per variant (88 RGB images). Room geometry hashes are invariant across each family's camera, material, detail, illumination and exposure interventions. People are disabled in this sweep. The receipt binds the actual native executable, source identity, target, feature flags and locked package versions. All cases are retained, including failures; no model inference filters pairs. The project page provides the complete matched annotation gallery and machine-readable receipt.

New captures export a full row-major pixel-space K, image dimensions, half-pixel centers, a versioned centered pinhole lens contract, normalized playback progress, eased trajectory progress and optional explicitly assigned seconds. Unassigned physical time remains unknown. Short handheld paths sample local metric translations and yaw/pitch/roll increments independently of a moving look-target, retaining swept collision checks. Requested high/low/zero proxy-overlap strata are selected before placement retries. Their all-ray proxy estimates are stored separately from rendered directed overlap over valid source pixels. A proxy-negative pair can retain a small rendered overlap. Lens distortion and sampled principal points are not implemented.

Table 4 reports means over directed rendered pairs, and the maximum per-view 99th-percentile self-reprojection error. There are 10 exact duplicate RGB images: the baseline/overlap interventions deliberately reuse the reference camera. These counts measure exact repetition in this matched study, not perceptual redundancy across a training population. Sensor white balance, blur, noise and optional JPEG are separate seeded RGB-only export transforms with individual operator regressions; they are not applied in this render sweep. This bounded check does not establish photographic realism, a calibrated sensor model, a fitted training mixture or downstream transfer.

Factor	Shared pixels (%)	Baseline (m)	Reproj. p99 max (px)
reference	58.5236	1.959	0.00008
portrait	45.4520	1.763	0.00017
wide baseline	41.5424	3.439	0.00008
handheld	64.7620	2.263	0.00009
low overlap	15.0775	3.518	0.00008
no proxy overlap	0.0000	1.648	0.00016
material seed	58.5236	1.959	0.00008
low detail	58.5236	1.959	0.00008
lighting seed	58.5236	1.959	0.00008
dim illumination	58.5236	1.959	0.00008
exposure	58.5236	1.959	0.00008

Table 4: Prespecified factor checks on matched room geometry. Directed overlap uses valid source pixels; the zero target refers to placement proxies.

4.5 Population distributions and publication contract

Room and object counts retain zero-instance rooms. Camera and photometric histograms state their own sample denominators. Placement heatmaps use 33 path samples per camera and normalize each panel independently. Feature shortcuts do not change population denominators. A seed's bit width and continuous parameter ranges do not establish ten-million-sample diversity; correlations, placement constraints and rendering affect distinguishable outcomes. Frozen SigLIP2 embeddings support separate redundancy audits, but no embedding-rank or downstream-learning result is claimed here.

The project page, paper, figures and calibration archives are built by a single Rust publication pipeline from one completed capture manifest. A compiled generator identity binds crate version, geometry grammar and source/dependency hashes. The publisher validates scene/camera/time identities, every requested channel, exact membership masks, annotation checks and population denominators before staging artifacts. The release/deployment gate checks input and output hashes, preserving explicitly separate historical reference studies. Author-written method text and measurements have distinct roles; publication attestation does not imply photographic or downstream validation.

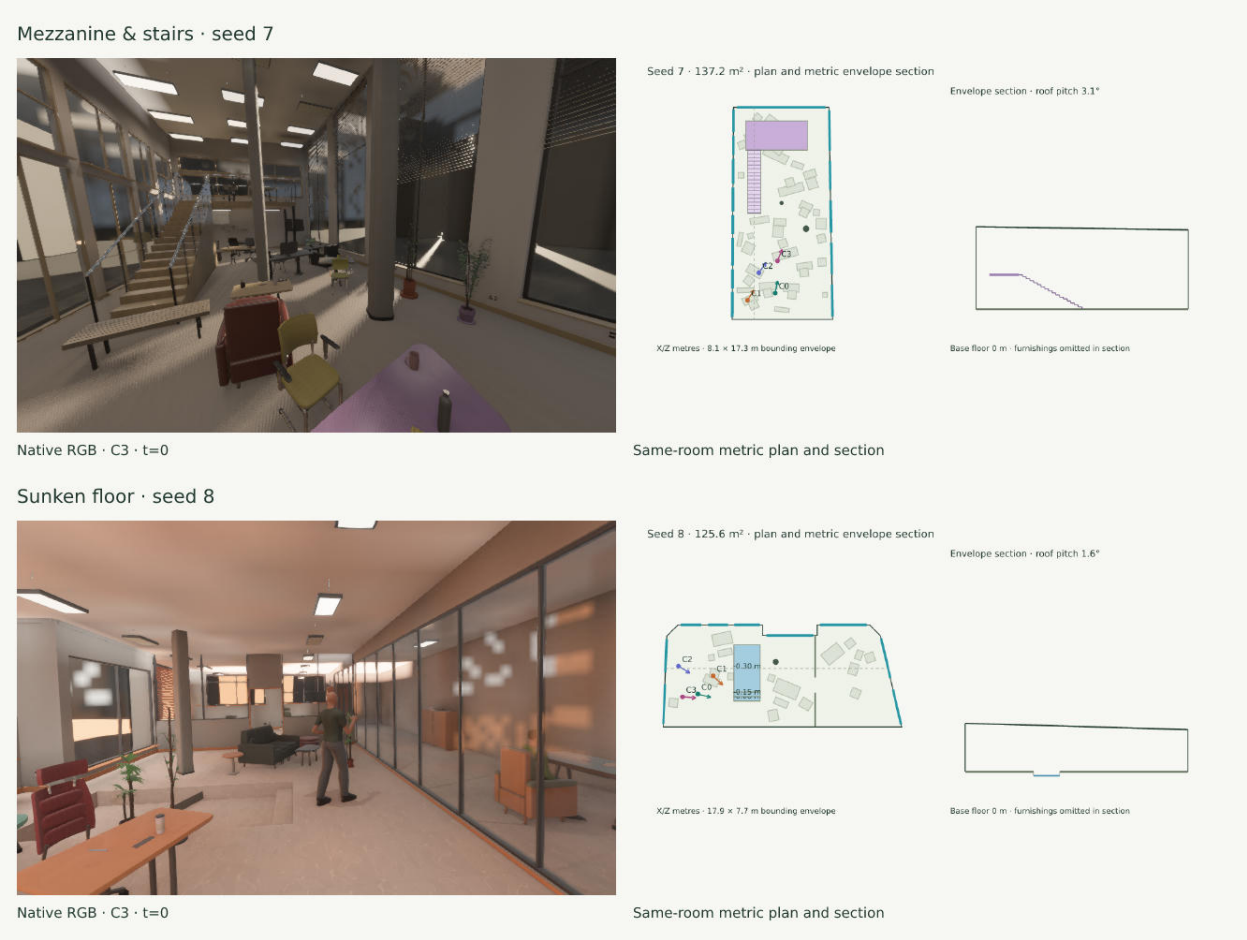


Figure 2: Native RGB and metric plans of structurally selected rooms. Raised and depressed levels, decks and stairs are generated geometry. Cyan marks glazing; orange/blue denote raised/sunken floors; violet denotes mezzanines and stairs.

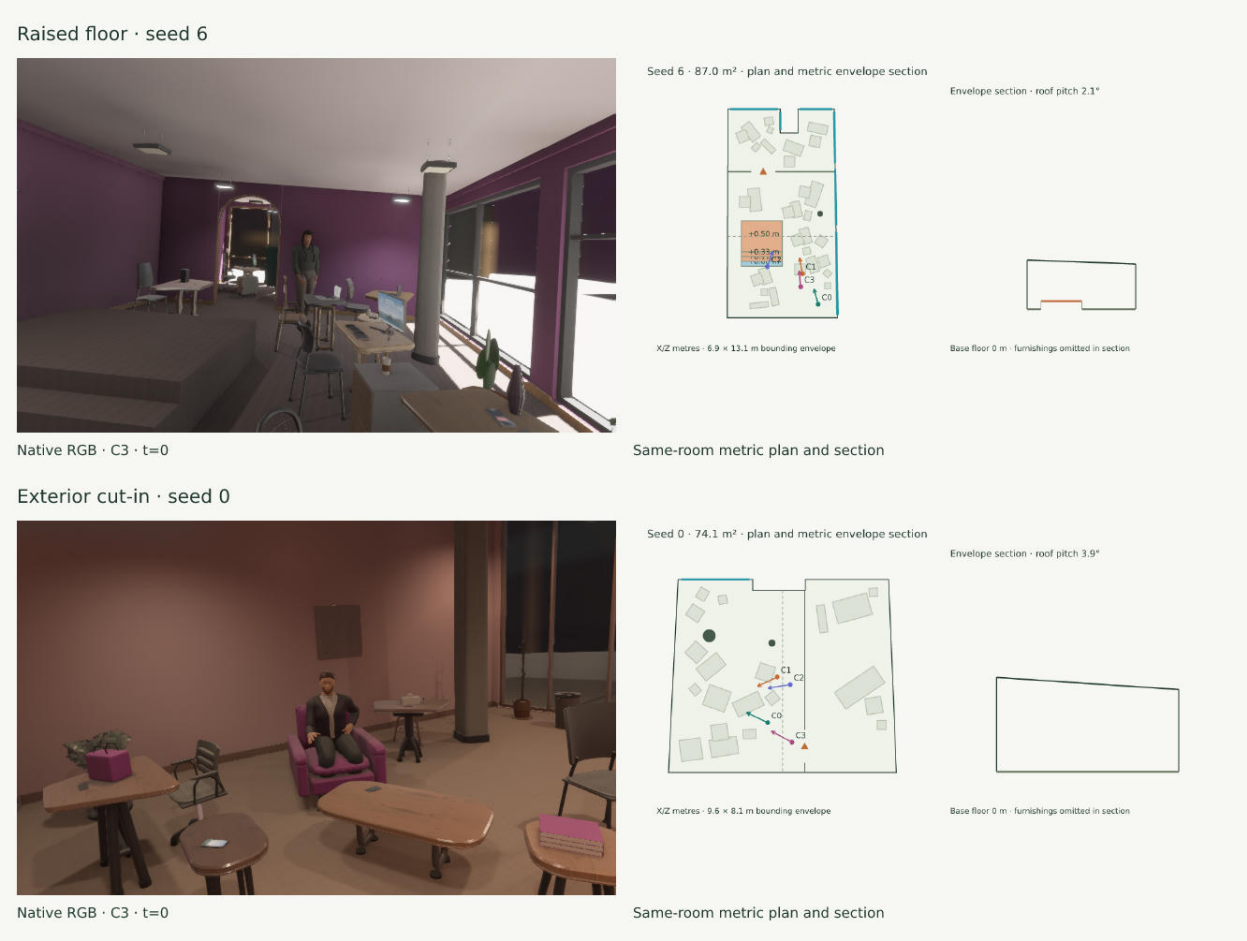


Figure 3: Native RGB paired with the corresponding envelope programs. Floor levels, portals, polygonal outlines and apertures vary; features can be outside a particular camera view.



Figure 4: Twelve metric envelope programs selected by architectural feature coverage from the full consecutive population. Plans are diagrams, not twelve additional RGB renders.

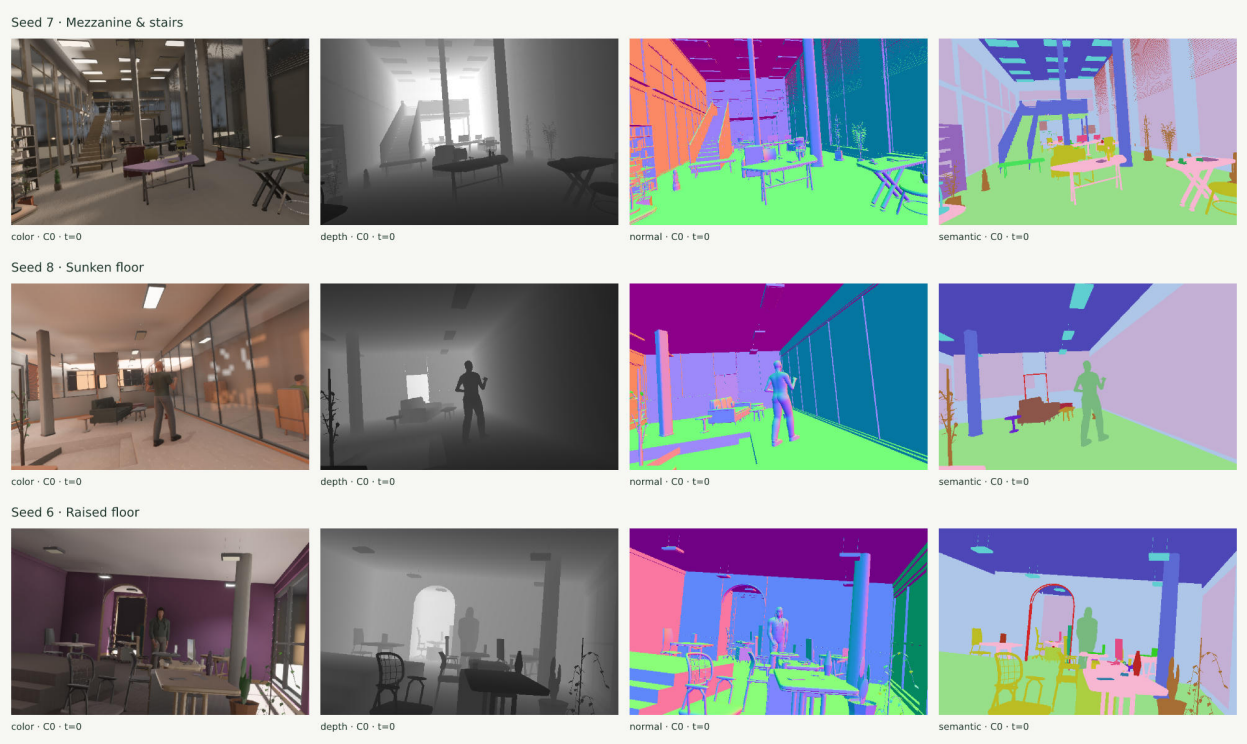


Figure 5: Matched RGB, depth, view-space normals and semantic display previews, camera 0 at t=0. All six annotations, camera identities and endpoint times remain accessible in the explorer.

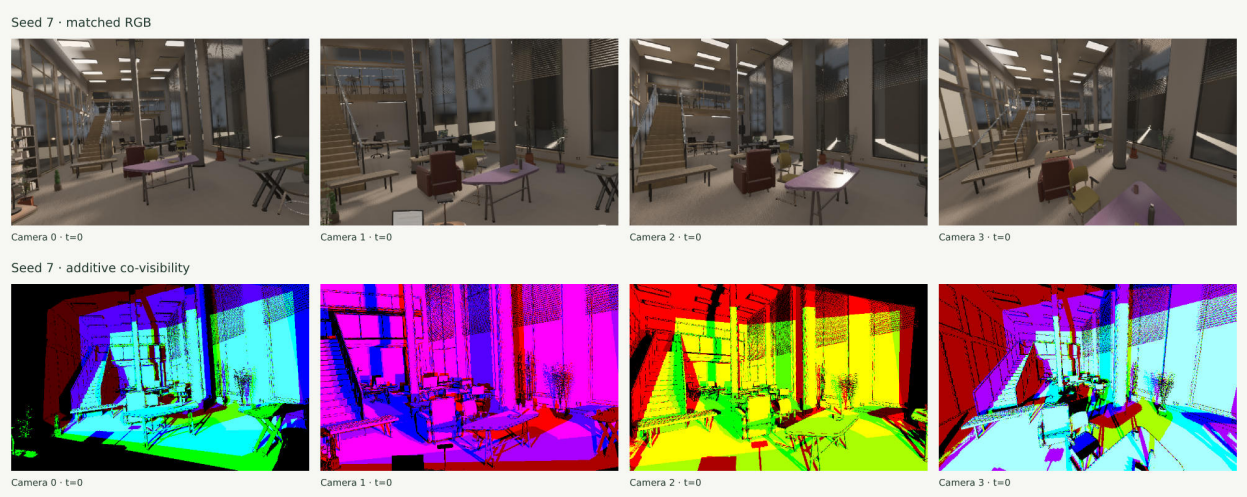


Figure 6: Seed 7 at t=0: four native RGB views above their production co-visibility annotations. Columns retain the same camera/time identity. All captured rooms expose this annotation in the same explorer.

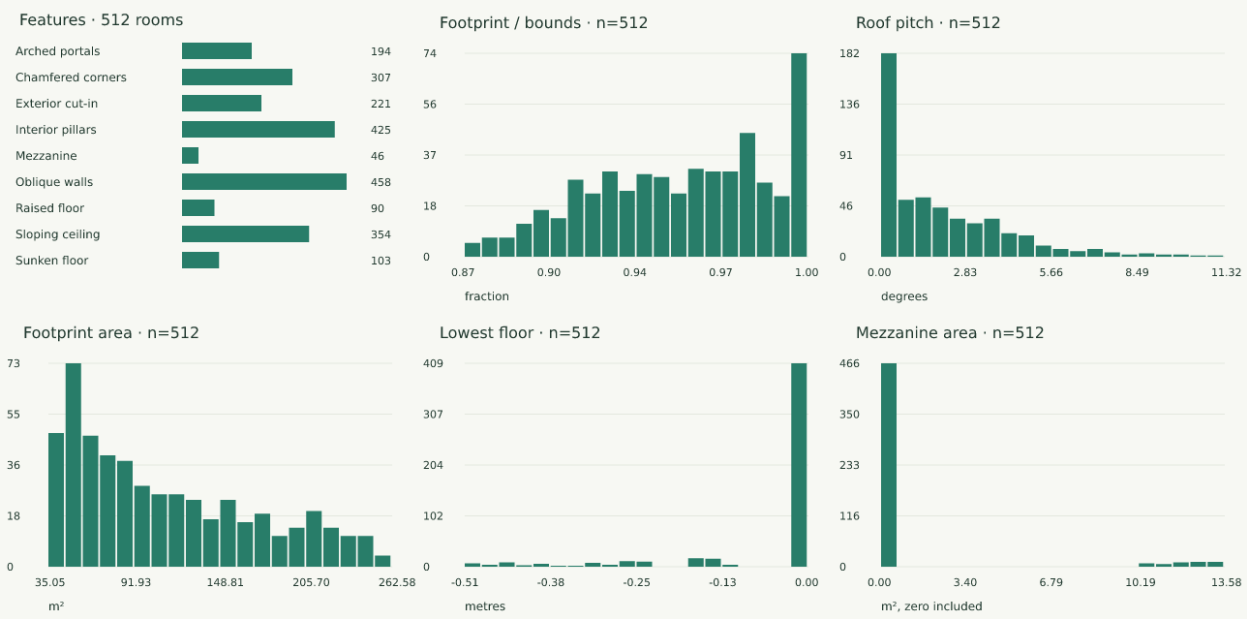


Figure 7: Architectural feature counts and continuous parameter distributions from the complete population; absent features are retained.

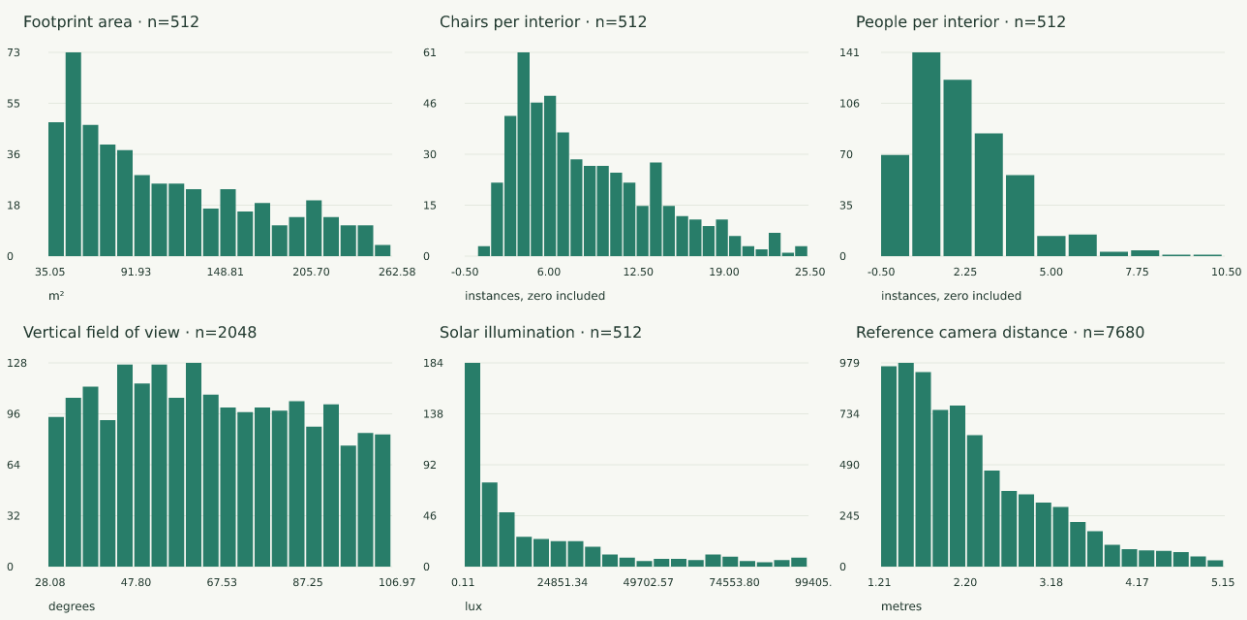


Figure 8: Furnishing, lighting and camera variation in the same population. Every panel states its denominator.

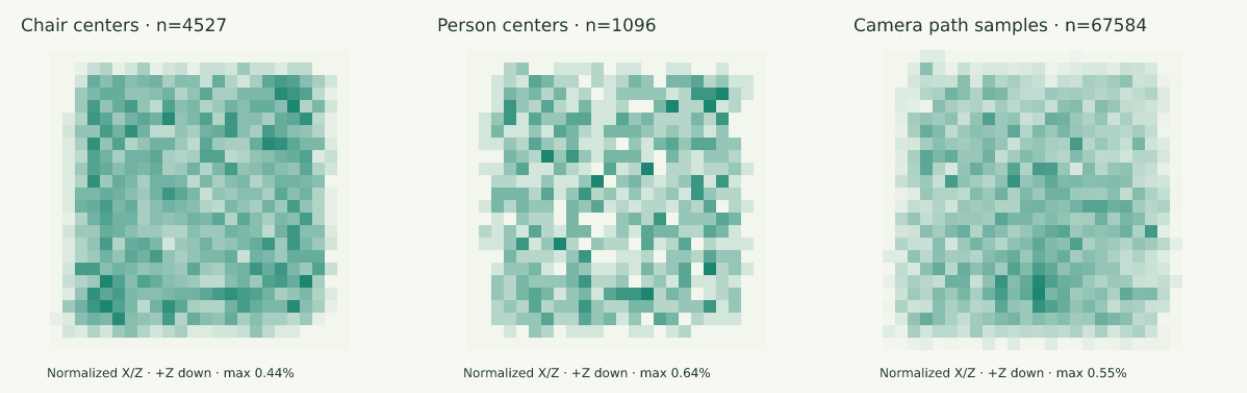


Figure 9: Chair centers, human centers and camera path samples in normalized room X/Z coordinates, with +Z down. Each panel reports its own sample count and color scale.

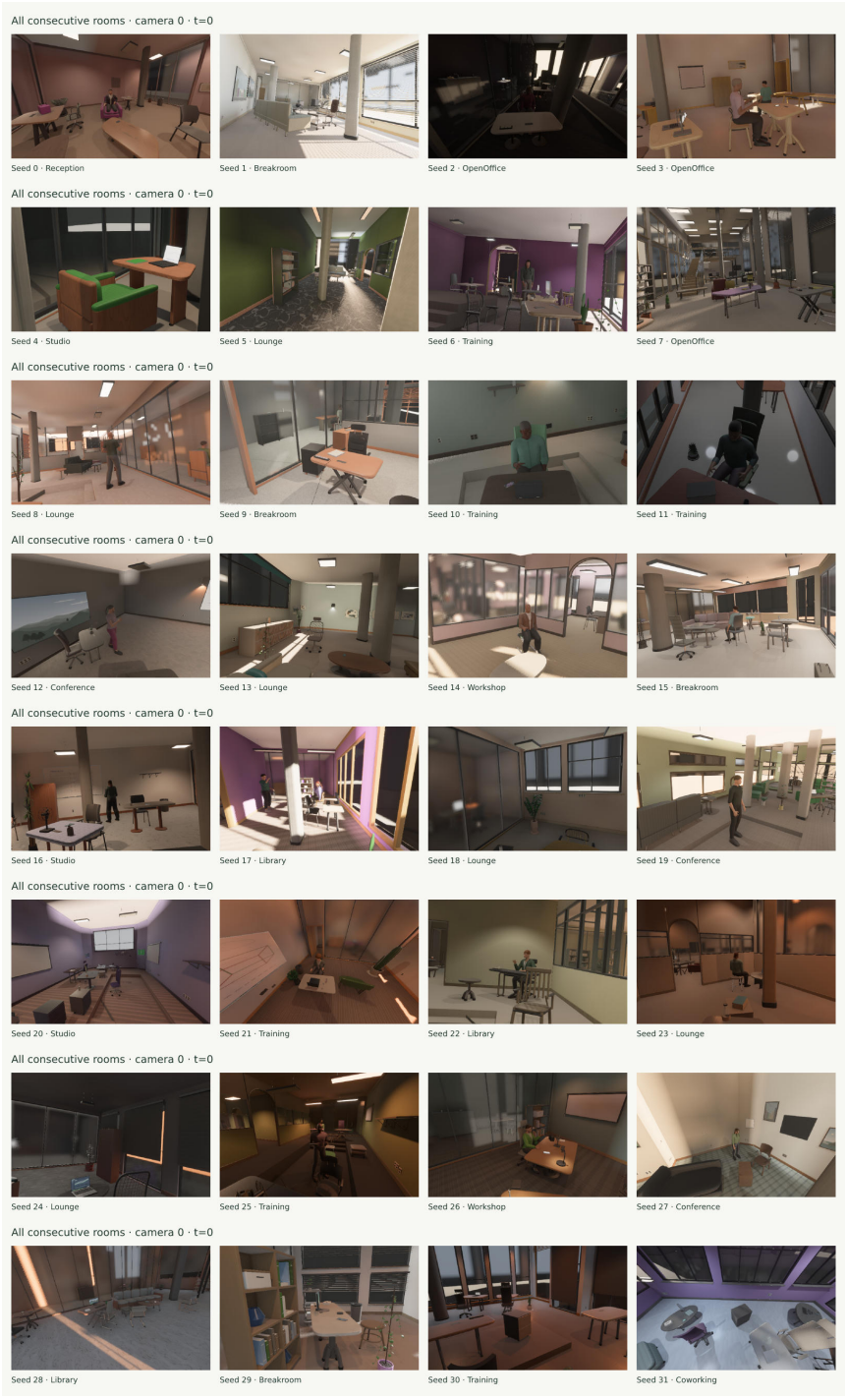


Figure 10: Every consecutively rendered room, camera 0 at t=0, including dark and sparsely furnished rooms. All four cameras, endpoints and six modes remain in the project-page explorer.

5 Limitations and remaining evidence

Procedural correctness, broad parameter variation and efficient capture do not establish photographic realism. Hair is a surface groom, skin lacks a full subsurface model, some furnishings remain sparse, object motion uses approximate collision proxies, and diffuse GI uses probes and simplified transport. Glass and lighting are approximations rather than full path tracing. Physical transport accuracy and visual plausibility require independent path-traced reference studies; the camera-baseline sweep does not measure either.

The current study is a generator audit and a small capture cohort, not a ten-million-scene run. It does not establish long-run resident-memory bounds, balanced coverage of all rare architectural combinations, collision-free arbitrary motion, suitability for every reconstruction loss, or transfer to real photographs. Shared-view defaults deliberately correlate cameras within a room; train/validation splits should therefore be made by room seed, not by image. A downstream assessment should freeze a training budget and real evaluation set, compare matched data quantities and camera policies, and measure reconstruction/calibration quality alongside redundancy. The manifests and metrics make that experiment reproducible without claiming its outcome in advance.

A Recorded camera-baseline reference study

The following baseline, population and gallery results retain their generator-21 capture identities and predate the architectural gallery in the main text. They measure the camera policy in their recorded scenes; they do not remeasure the architectural cohort above. Both studies report absolute measurements with separate denominators. The camera-baseline reference build (generator 21) was audited on 2,048 consecutive layout seeds (24000–26047) at furnishing density 0.65, static human density 0.25, four cameras and default baseline $b = 0.5$. All layouts pass geometry, swept collision and sampled camera constraints. The population contains 508 numeric series, 406 with nonzero observed variance. These are correlated measurements.

The rendered population contains 512 distinct consecutive rooms (24000–24511), four 320×240 cameras and times $0, 0.5, 1$. The first 128 rooms are additionally captured at $b = 0, 0.25, 0.75, 1$, yielding 1,024 room/configuration captures and 12,288 views. Every row in the parameter sweep uses the same 128 rooms, with identical geometry, people, materials and lighting, verified by manifest hashes. No failed, dark, or low-overlap sample is replaced. Native Auto lighting and shadows are enabled; baked diffuse GI is disabled for the sweep. The selected gallery uses a separate full-lighting recipe. Captures use the checkout-local wgpu command-cache/upload optimizations; published crates resolve registry wgpu. Timing observations do not qualify unpatched package performance.

Baseline	Distance (m)	Overlap (%)	Shared (%)	All three (%)	Proxy misses
0.00	0.19	72.8	92.1	61.0	177 / 1152
0.25	1.33	55.2	85.2	39.9	96 / 1152
0.50	2.19	46.5	80.7	28.4	52 / 1152
0.75	2.94	38.9	76.9	22.0	22 / 1152
1.00	3.60	32.9	72.8	17.3	11 / 1152

Table 5: Absolute measurements at five settings of the camera-baseline program. Distance is the median room-mean reference distance, measured at 33 times. Overlap averages bidirectional reference pairs at three rendered times. Shared and all-three fractions use valid source-pixel observations. Each row contains 128 rooms, 1,536 views and 1,152 reference pairs; its proxy threshold depends on baseline.

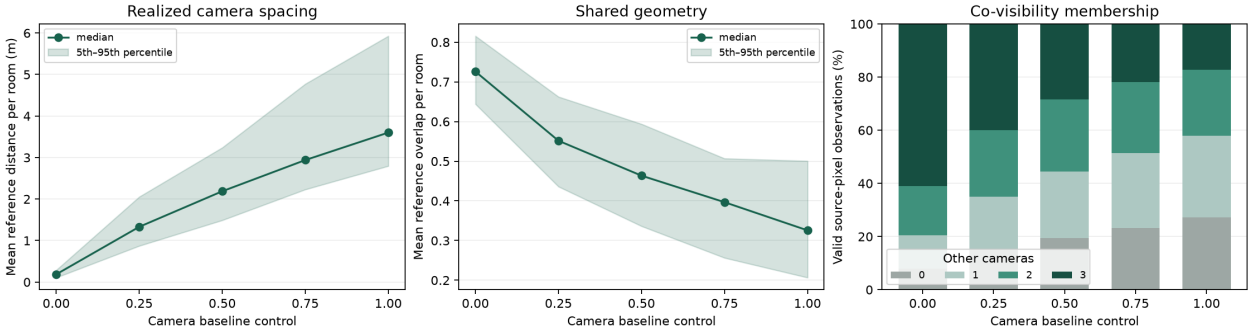


Figure 11: Realized spacing, room-level overlap quantiles and co-visibility cardinality versus the camera-baseline parameter. Pixel observations and views from the same room are correlated.

At default baseline, all 512 rooms contribute 6,144 views and 4,608 reference-pair/time observations. Mean reference overlap is 45.7%, with minimum 16.0%; 210 pairs miss the proxy target. 80.3% of valid source-pixel observations are visible to another camera, and 27.9% to all three others. The denominator is 471,859,200 valid pixel observations, not unique 3D points. Membership excludes the source camera.

The independent diagnostic requires nearest-target-pixel depth agreement within $0.01\,\mathrm{m} + 0.002z$. The production GPU annotation uses a separate symmetric tangent-plane predicate; its exact masks are exported with the gallery. Both use the first geometric surface, treating glass as annotation-opaque. Reflections and refractions are not counted as geometric correspondence.

There are 0 rooms with a view containing at most two semantic classes, and 0 of 357 occupied primary rooms have no person pixels in any captured view. Maximum per-view p99 depth/position disagreement is 0.0000038 m. These are annotation and coverage checks, not photographic-realism metrics. Complete distributions, source identities, configurations and commands are in `docs/camera_baseline_v21.md`. Bounded capture processes do not establish unlimited-process memory stability.

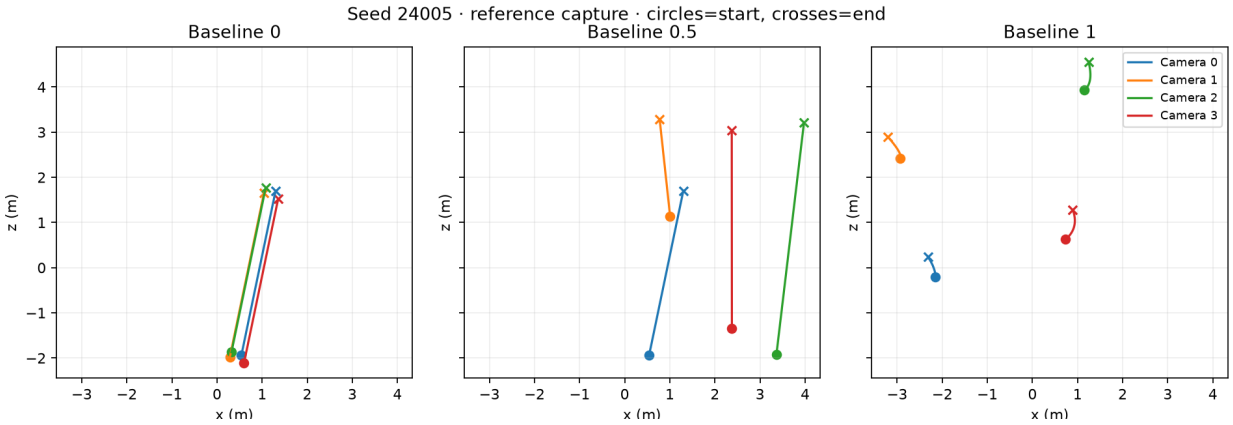


Figure 12: Camera programs for one fixed reference room at three baseline settings, using common metric axes.

A.1 Seeing the baseline control in a fixed room

Figures 13 and 14 show the visual effect of camera-baseline sampling in two fixed scenes. Each column contains all four synchronized RGB views and a top-down camera plan at $t = 0$. Columns change only the camera policy: geometry, furnishings, static people, materials and lighting are identical within a room, verified by hashing all noncamera manifest fields. The project page's reference gallery provides five captured settings ($b = 0, 0.25, 0.5, 0.75, 1$), an interactive four-view comparison, and a shared-surface overlay backed by exact exported GPU masks. The figures select three of these settings for legibility; they do not select or crop individual camera views.

These selected illustrations use native Auto lighting, 1024 diffuse-GI rays per probe and 768×480 images. They are separate from the consecutive population experiment above. The ten room/configuration captures yield 40 RGB views at one time; the two default groups reuse the matched gallery's $t = 0$ exports. Image conversion uses a fixed sRGB transfer after renderer tone mapping, with no per-image exposure normalization.

Camera indices identify members of a resampled group, not fixed physical viewpoints across columns. The reference pose and sampled intrinsics may change as rejection finds a feasible group. Hence the examples illustrate the complete baseline policy, not translation of a fixed calibrated rig. Plans retain identical metric axes within each room and show furniture footprints as proxies; short wedges indicate viewing direction, not full frusta or occlusion. Column headers give the mean camera-0 distance at this timestep and the fraction of valid source-pixel observations shared with at least one peer, pooled across all four views. The latter uses the production GPU surface test, distinct from the population depth diagnostic. Pixel observations repeat surfaces; the default glass policy retains the first transmissive surface. An opt-in through-glass policy excludes specular-transmissive surfaces consistently from depth, normal, position, semantic, optical-flow and co-visibility hits, following unrefracted rays; RGB transmission remains unchanged.

A.2 Diversity diagnostics

The population export retains zero-preserving object counts, parameter distributions, camera intrinsics and extrinsics, placement heatmaps, correlations and occupancy signatures. These are correlated projections of the procedural program, not independent degrees of freedom. For categorical frequencies p, the entropy-effective count is $\exp(-\sum_j p_j \log p_j)$. The occupancy signature uses a 12-by-12 object-category grid and quantized partition geometry, excluding random seeds and textures. Its HyperLogLog estimate is a bounded geometric redundancy diagnostic, not proof of ten million useful examples.

A seed's bit width does not establish dataset diversity. Continuous parameter ranges have no useful finite cardinality before quantization; topology, placement constraints, correlations and rendering reduce distinguishable outcomes. Optional frozen SigLIP2 embeddings [3] support image-space redundancy audits, with scene-level aggregation and cross-room nearest neighbors. No embedding-rank or downstream-learning claim is made for this camera-baseline sweep.

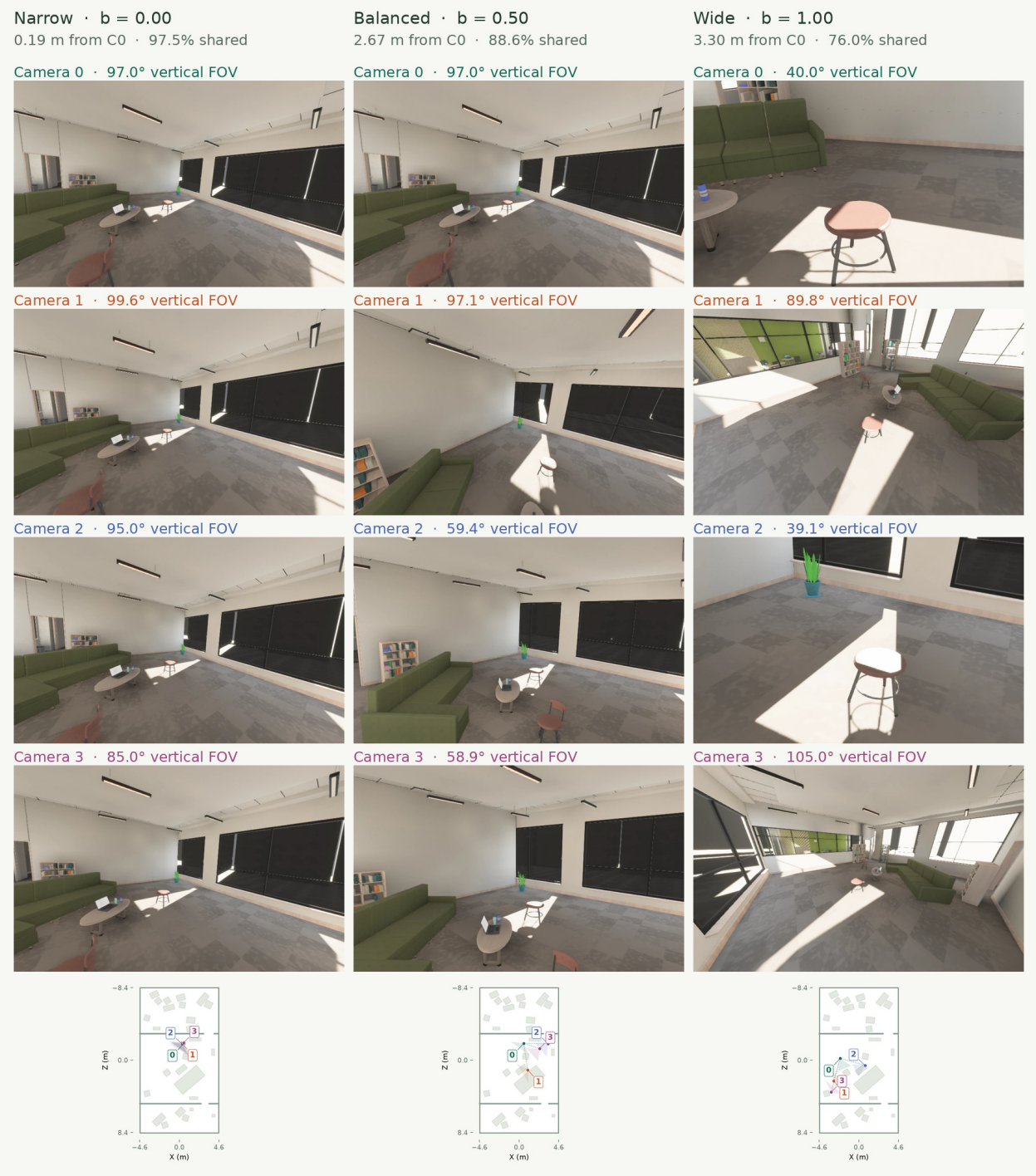


Figure 13: Fixed sunlit seating scene (seed 24005), with narrow, balanced and wide baseline policies from left to right. Rows retain cameras 0–3 in order; the bottom row locates their centers in the same primary room. Close views concentrate on similar content; wider sampling changes perspective and the surfaces jointly observed. Header values describe these particular captures, not population averages.



Figure 14: Fixed workspace scene (seed 24000), using the same column and camera ordering as Figure 13. The room and its lighting are held fixed across baseline values. Feasibility depends on furniture, partitions, camera intrinsics and overlap constraints, so the control is not a requested distance in metres and individual realizations need not vary monotonically.

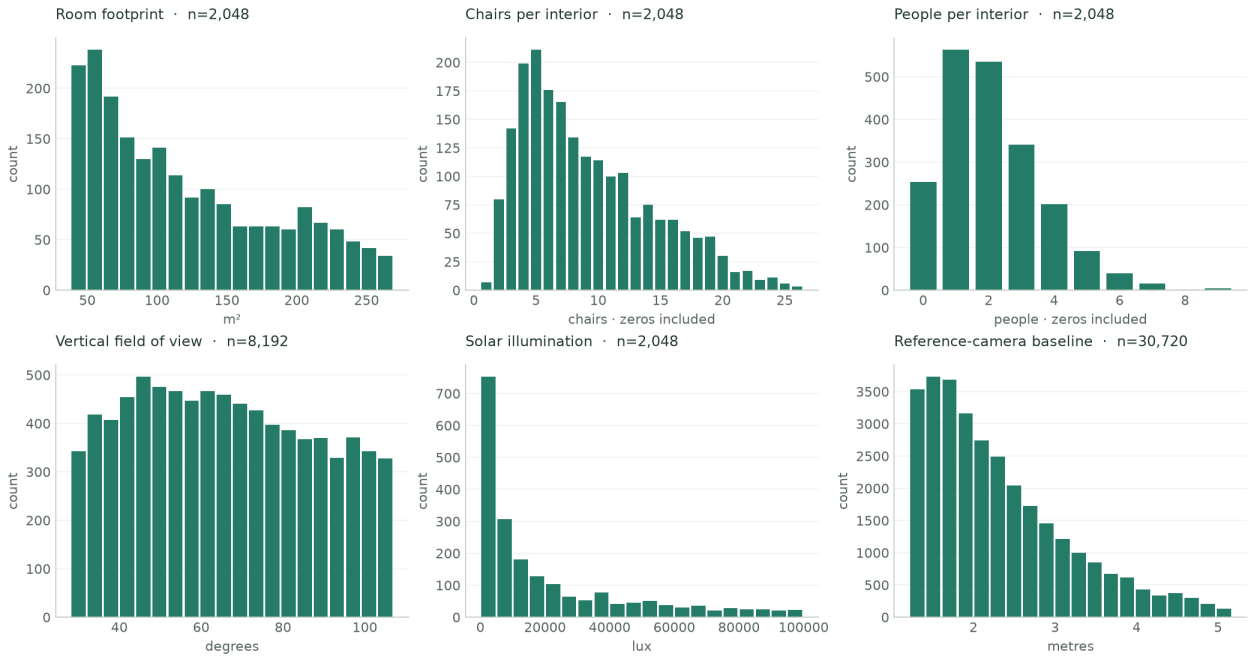


Figure 15: Absolute distributions in the camera-baseline reference population of 2,048 rooms at the default camera baseline. Each panel preserves its denominator and units.

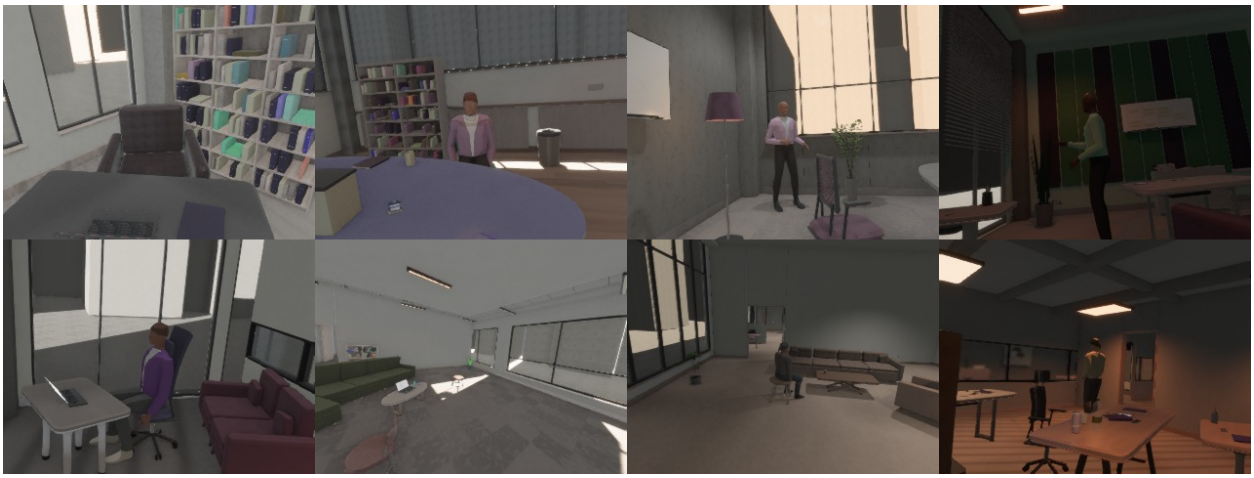


Figure 16: Camera-zero RGB for the first eight consecutive rendered rooms at the default baseline, without aesthetic filtering.

B Matched gallery and viewer

The accompanying project page links a separate reference gallery, preserving these earlier captures without presenting them as current architectural examples. Two example rooms (seeds 24005 and 24000) were captured at 768×480, with four synchronized cameras, three normalized trajectory times and all eight render modes. All mode switches reuse these exact scene/camera/time identities. The native CLI used Auto quality, static human density 0.25 and 1024 indirect-lighting rays per probe. Depth/position reconstruction and camera reprojection checks validate alignment before generating website previews. Display mappings for depth, normals, positions and temporal vectors are declared alongside the visualizations; they do not replace the lossless numeric exports.

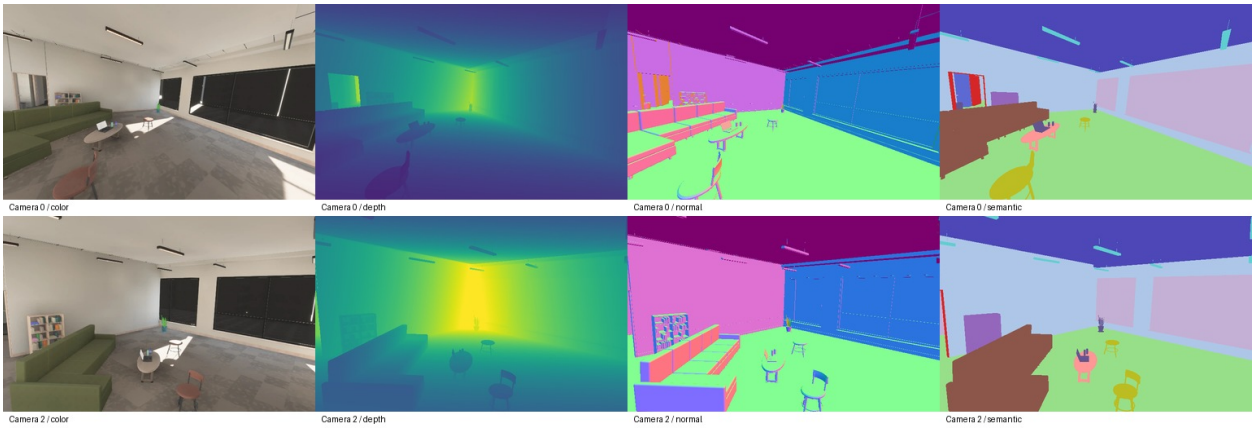


Figure 17: Selected reference room, seed 24005: RGB, metric-depth visualization, view-space normals and semantic labels for cameras 0 and 2 at the same time. Camera selection and pixel-aligned annotation comparisons are interactive on the project page. These illustrations are separate from the consecutive population evaluation.

B.1 Co-visibility renderings and camera-specific supervision

Figures 18 and 19 show the production GPU annotation for the matched reference gallery. These are actual exported surface-membership masks at the same camera poses, intrinsics and time as the RGB images. They are distinct from the looser depth-based population diagnostic in the baseline table. The figures use the existing selected captures; they add no rooms to the population denominator.

Co-visibility shares the float32 surface raster used by the other geometric annotations. For source camera s, pixel p and time t, the exported unsigned 16-bit integer is

$$M_s(p,t) = \sum_{\substack{j=0 \\ j \neq s}}^{N-1} 2^j V_{s \to j}(p,t), \qquad 1 \leq N \leq 16,$$

where $V_{s \to j} \in \{0,1\}$ records visibility of the source surface in peer camera j. The source camera's own bit is always zero; the editor has no membership bit. The ordered camera legend remains identical across views and timesteps in a sample. A peer's binary supervision is recovered by testing $(M_s \,\&\, 2^j) \neq 0$; the number of observing peers is $\mathrm{popcount}(M_s)$. Thus the integer preserves camera identities, unlike a shared-pixel count alone.

The GPU pass projects a valid source surface into each peer, checks near/far clipping and image bounds, and reads the nearest target pixel. For source point x, target point y, corresponding world-space unit normals n_s, n_j and target axial depth d, the surface-agreement test is

$$\max(|(y-x)\cdot n_s|, |(y-x)\cdot n_j|) \leq \max(0.001\,\mathrm{m}, 10^{-4} d).$$

This local-plane test avoids comparing depths along different pixel-center rays on sloped surfaces. It remains resolution- and tolerance-dependent: silhouettes, thin geometry, curved surfaces and near-contact surfaces can be ambiguous. Pixel-level membership need not be symmetric after raster reprojection. Glass is the first opaque geometric surface, so reflected, refracted and transmitted RGB content does not imply co-visible geometry.

The additive RGB8 preview sums the contributions of set camera bits without tone mapping. For four cameras, the shared legend is $C_0 = (170,0,0)$, $C_1 = (0,255,0)$, $C_2 = (0,0,255)$ and $C_3 = (85,0,0)$. For example, $M = 10 = \texttt{0b1010}$ means peers 1 and 3, with color $(85,255,0)$. These colors encode camera membership, not semantic classes or overlap magnitude. The palette adapts deterministically to camera count and remains uniquely decodable for all supported combinations in the original lossless PNG.

A separate uint8 source-valid mask A_s distinguishes an unshared surface ($A_s = 1, M_s = 0$) from background ($A_s = 0, M_s = 0$); both appear black in the additive preview. Folder exports retain membership and validity as $[H, W, 1]$ arrays in NPZ. Chunked exports use $[B, T, N, H, W, 1]$. Training should consume these numeric arrays and the camera metadata, rather than rasterizing the PDF or decoding a resized or lossy preview. Shared percentages below divide by valid source-pixel observations, including unshared surfaces, not unique 3D points.

All eight $t = 0$ gallery annotations used for this check round-trip exactly between exported uint16 masks and their original additive PNGs. Source validity, self-bit exclusion, shared legends, capture identities and per-peer/count

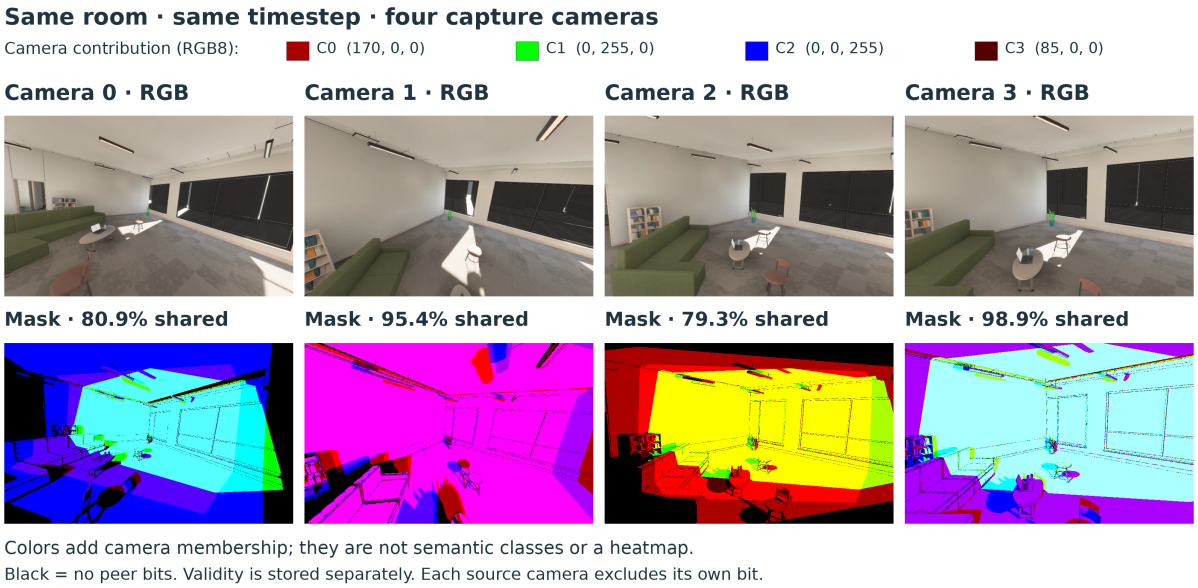


Figure 18: Matched RGB (top) and production co-visibility (bottom) for all four cameras in seed 24005 at $t = 0$, from native 768×480 captures. Columns retain the exact viewpoint and full image extent. Camera-bit colors share one legend; each source excludes its own bit. Percentages count valid source pixels visible in at least one peer, for these captures only.

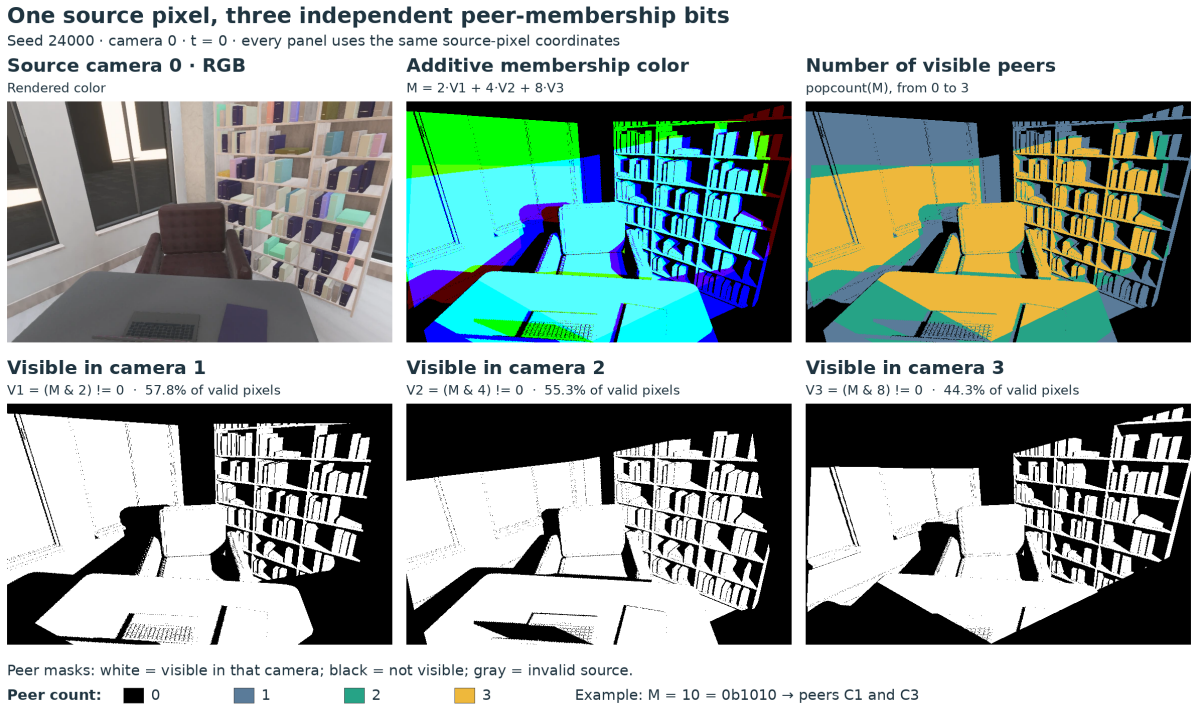


Figure 19: Camera-specific supervision: seed 24000, source camera 0, $t = 0$. Top: RGB, additive code and derived peer count. Bottom: independent peer bit tests, all in source-pixel coordinates. White means visible, black means not visible; invalid source pixels would be gray. The peer masks can overlap: their weighted sum reconstructs membership, and their unweighted sum gives the count. The count palette is separate from the camera legend.

denominators are checked before composing the figures. Original capture hashes and depth/position reprojection are checked separately. The reproducible builder and per-view evidence are `scripts/build_paper_covisibility.py` and `docs/evidence/paper_co_visibility_v21/report.json`.

B.2 Human motion illustration

Human clothing uses body-derived cross-sections, localized folds and fitted garment details. Viewer pose joints can show through people while environment depth occludes them; editor projections persist through regeneration. ARDY controls expose diffusion steps, history length, guidance and path conditioning. The reference page's separately identified motion example contains two accepted trajectories, two cameras and 120 frames at the model's 20 Hz rate. It is an illustration of the optional motion pipeline, not part of the static camera-baseline experiment or a motion-quality benchmark.

References

[1] D. Xie et al. LRM-Zero: Training Large Reconstruction Models with Synthesized Data. NeurIPS, 2024. https://arxiv.org/abs/2406.09371.

[2] Y. Chen et al. MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images. ECCV, 2024. https://arxiv.org/abs/2403.14627.

[3] M. Tschannen et al. SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. 2025. https://arxiv.org/abs/2502.14786.